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# Machine Learning Operations

## Case Study

Question 2: Using Chapter 5's "Data Access Before Validation and Launch to Production," analyze the impact of missing features on model reliability. How should MedInsure implement schema validation before deployment?

## Team Members:

|                           |                   |
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# Impact of Missing Features on Model Reliability

- Missing features (like **hospital\_risk\_score**) break the assumptions under which the model was trained.
- Degrades accuracy and makes the model behave unexpectedly. (Model may incorrectly assign risk scores → poor claim decisions.)
- Missing features shift the **data distribution**.
- Leads to **loss of trust**, inconsistent predictions across environments, and operational failures.



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# Freeze & Bundle Critical Data with the Model

- Freeze all lookup tables used during training. e.g., **hospital\_risk\_score maps, diagnosis-category groups, fraud-flag mappings**
- Bundle them inside the model artifact
- Ensures model uses the **exact same reference data** in dev, staging, and production
- Prevents discrepancies when pipelines or databases change
- Guarantees **reproducibility of predictions**, even years later.



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# When the Model Needs External Data Lookup at Scoring Time

- Some features cannot be bundled because they must be **fresh, dynamic, or large**. e.g., latest fraud watchlist, updated hospital rating feeds, policyholder risk-profile changes
- Validate all external data **before** the model consumes it. Check schema, datatype, missing fields, invalid codes



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# In general,

Freeze **static** reference data with the model. Validate and control **dynamic** external data. Ensure all external lookups are versioned, logged, and monitored

This protects against:

- Missing features
- Schema drift
- Silent failures
- Inconsistent scoring across environments
- Attackers manipulating external fields